

Module 3 — Read 1

One categorical, one numerical, numerical by group

Module 3, Read 1: Tools for Describing One Variable

This reading is your self-paced textbook for Module 3, Part 1. Work the three sections below at your own pace; use Check yourself boxes before **Quiz A**. You will build the descriptive toolbox for one categorical, one numerical, and then move a pair of variables: one numerical variable among the categories of one categorical variable.

Learning outcomes

Work through Sections 1–3 below. Each ends with Check yourself questions.

After this reading you should be able to do the following.

Section 1: One categorical variable

One categorical (including binary) — counts, proportions, bar charts

- Classify variables as numerical or categorical (and binary when there are two levels).
- Summarize one categorical variable with counts and proportions using x , n , and $\hat{p} = x/n$.
- Draw bar charts (and recognize pie charts) in ggplot2; use `count()` and `mutate(prop = ...)`.

Section 2: One numerical variable

One numerical — center, spread, shape, graphs, and boxplots

- Compute mean and median by hand and with `summarize()`; choose between them by purpose, either a typical member's experience (median) vs. a total shared equally across units (mean).
- Graph one numerical variable with histograms and dot plots; describe center, spread, shape, and unusual observations.
- Compute range, SD, MAD, quartiles, IQR, and the five-number summary by hand; interpret boxplots.

Section 3: One numerical and one categorical

Numerical outcome across groups — grouped summaries and boxplots

- Compare spread across groups with SD, IQR, and side-by-side boxplots.
- Use `group_by()` with `summarize()` for descriptives and ggplot2 to compare groups with histograms, dot plots, or boxplots.
- Describe the Welch two-sample t -test and run `t.test(y ~ group)`.

Course Path

Module	Focus	Typical question
1	Inferential — one binary outcome	Is the population proportion p different from a claimed p_0 ?
2	Inferential — one proportion or mean from a sample	What is a plausible value for p or μ in the population ?
3	Descriptive — several variables in your dataset	What patterns appear in this sample ? (No required generalization to a larger population.)

Descriptive statistics summarize and visualize the data as they appear in your file. You still use the same statistic notation from earlier modules (x , n , $\hat{p}$, $\bar{x}$, s), but the goal is to describe this dataset, not to test or estimate a population parameter such as p or μ .

Section 1: One categorical variable

i Start with one variable

When you have one categorical variable, list how many observations fall in each level (category). Report counts and proportions (or percentages). The natural graph is a bar chart; a pie chart shows the same information as wedge areas.

i Supplemental videos — bar plots

Resource	Type
Creating bar graphs — Khan Academy	Video
Reading bar graphs — Khan Academy	Video

Variable types

A variable is a characteristic that may differ from one observational unit to the next.

Type	Definition	Examples
Categorical (qualitative)	Levels are labels; order or distance between labels is not meaningful for arithmetic	Species, island, sex (M/F), yes/no
Numerical (quantitative)	Values are counts or measurements where order and distance matter	Flipper length (mm), body mass (g), age (years)

Binary categorical variables have exactly two levels (yes/no, male/female, pass/fail). They are categorical, not numerical.

Steps for one categorical variable:

1. Count how many observations are in each level.
2. Convert counts to proportions (or percentages) that sum to 1 (or 100%).
3. Draw a bar chart (or pie chart) with clear axis/legend labels.

Notation — same as Modules 1 and 2

For a categorical variable with k levels, let x be the count in one level and n the total number of observations (non-missing).

Symbol	Meaning
n	Sample size (number of observational units)
x	Count in one category

Symbol	Meaning
$\hat{p} = x/n$	Sample proportion in that category (a statistic)

If you report a percentage, multiply by 100: $100 \times \hat{p}\%$.

Example (binary). Of $n = 25$ students, $x = 18$ prefer online homework.

- Count: $x = 18$
- Proportion: $\hat{p} = 18/25 = 0.72$
- Percentage: 72%

Example (multi-level). Penguin species in a dataset: Adelie $x_1 = 152$, Chinstrap $x_2 = 68$, Gentoo $x_3 = 124$, with $n = 344$ total.

- Adelie proportion: $\hat{p}_{\text{Adelie}} = 152/344 \approx 0.44$
- Each level gets its own count and proportion; proportions across all levels sum to 1.

Graphs

For one categorical variable, the natural displays are bar charts (counts) or pie charts (proportions as wedges). Numeric summaries are counts x and proportions $\hat{p} = x/n$ for each level.

Goal	Numeric summary	Graph
How many in each level?	Counts x per level	Bar chart of counts
What fraction in each level?	Proportions $\hat{p} = x/n$ (or %)	Bar chart or pie chart
What is the most common level?	Level with largest count / proportion	Bar chart or pie chart

In Project 3, you'll classify each of your four variables. For every categorical variable, include at least one count/proportion and one bar chart in your code notebook.

💡 Interpret the output — code comes in the demo lab

The chunks below use tidyverse to produce tables and figures. Focus on reading counts, proportions, summaries, and graphs. You will learn how to write these steps in [Demo 1](#) and [Demo 2](#).

Example 1 — Penguin species

```
library(tidyverse)
if (!requireNamespace("palmerpenguins", quietly = TRUE)) {
  install.packages("palmerpenguins", repos = "https://cloud.r-project.org")
}
library(palmerpenguins)

penguins |>
  count(species) |>
  mutate(prop = n / sum(n), pct = round(100 * prop, 1))

# A tibble: 3 x 4
  species      n prop  pct
<fct>    <int> <dbl> <dbl>
1 Adelie    152  0.442  44.2
```

```
2 Chinstrap    68 0.198 19.8
3 Gentoo      124 0.360 36
```

```
ggplot(penguins, aes(x = species)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Penguin species", x = "Species", y = "Count")
```

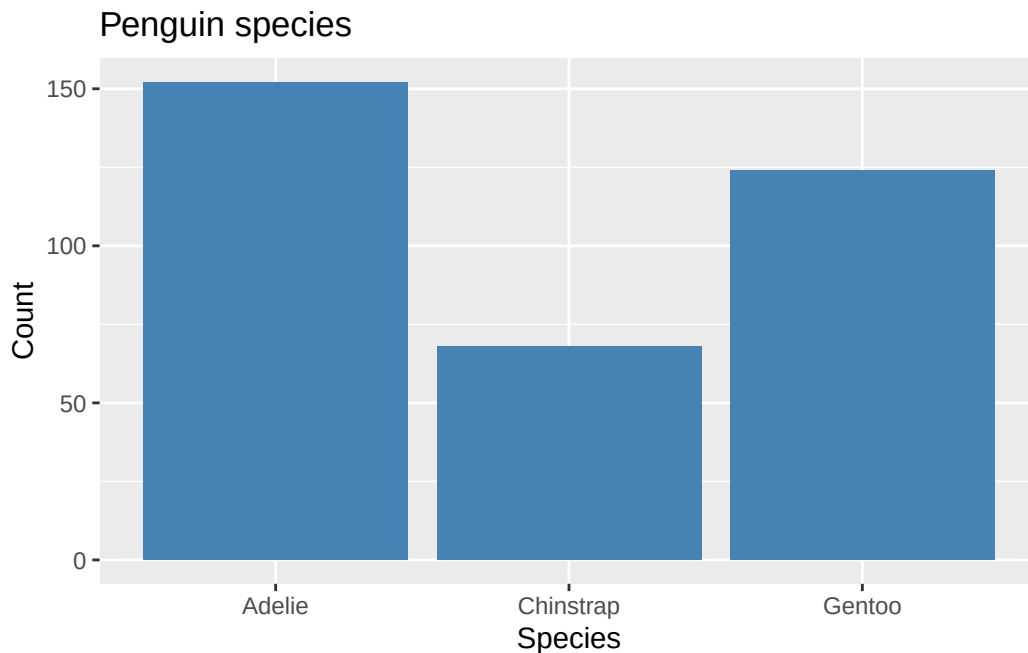


Figure 1: Counts of penguin species in the Palmer penguins sample.

Interpret: Which species is most common? Report one count x , total n , and proportion $\hat{p} = x/n$ for that level.

Answer: Adelie is most common with $x = 152$ out of $n = 344$, so $\hat{p} = 152/344 \approx 0.44$ (about 44%).

Example 2 — Vehicle class

```
data("mpg", package = "ggplot2")
mpg |> count(class) |> mutate(prop = n / sum(n))
```

```
# A tibble: 7 x 3
  class      n  prop
  <chr>    <int> <dbl>
1 2seater     5 0.0214
2 compact    47 0.201
3 midsize    41 0.175
4 minivan    11 0.0470
5 pickup     33 0.141
6 subcompact 35 0.150
7 suv        62 0.265
```

```
ggplot(mpg, aes(x = class)) +
  geom_bar(fill = "darkorange") +
```

```
labs(title = "Vehicle class counts", x = "Class", y = "Count") +
theme(axis.text.x = element_text(angle = 35, hjust = 1))
```

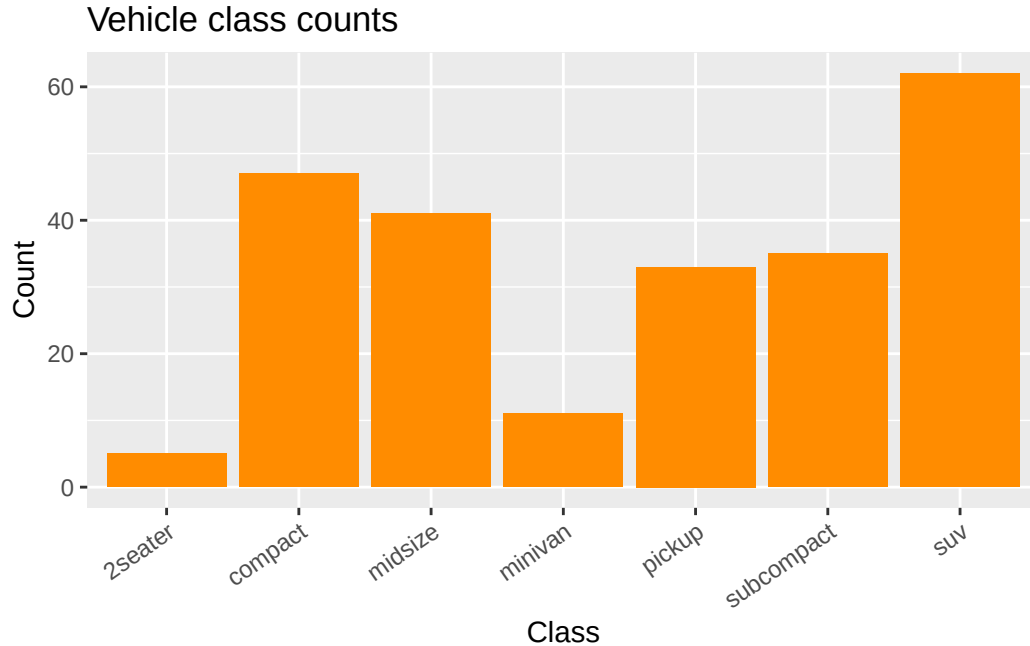


Figure 2: Number of car models by vehicle class.

Interpret: How many suv models appear? What fraction of all models is compact?

Answer: 62 suv models appear. Compact models are $47/234 \approx 0.20$ (about 20%) of all $n = 234$ models.

Example 3 — Binary categorical

```
set.seed(303)
nest <- tibble(success = sample(c("yes", "no"), 48, replace = TRUE, prob = c(0.62, 0.38)))
nest |> count(success) |> mutate(prop = n / sum(n))
```

```
# A tibble: 2 x 3
  success     n prop
  <chr>   <int> <dbl>
1 no       19 0.396
2 yes      29 0.604
```

```
ggplot(nest, aes(x = success)) +
  geom_bar(fill = "seagreen") +
  labs(title = "Nest success", x = "Success?", y = "Count")
```

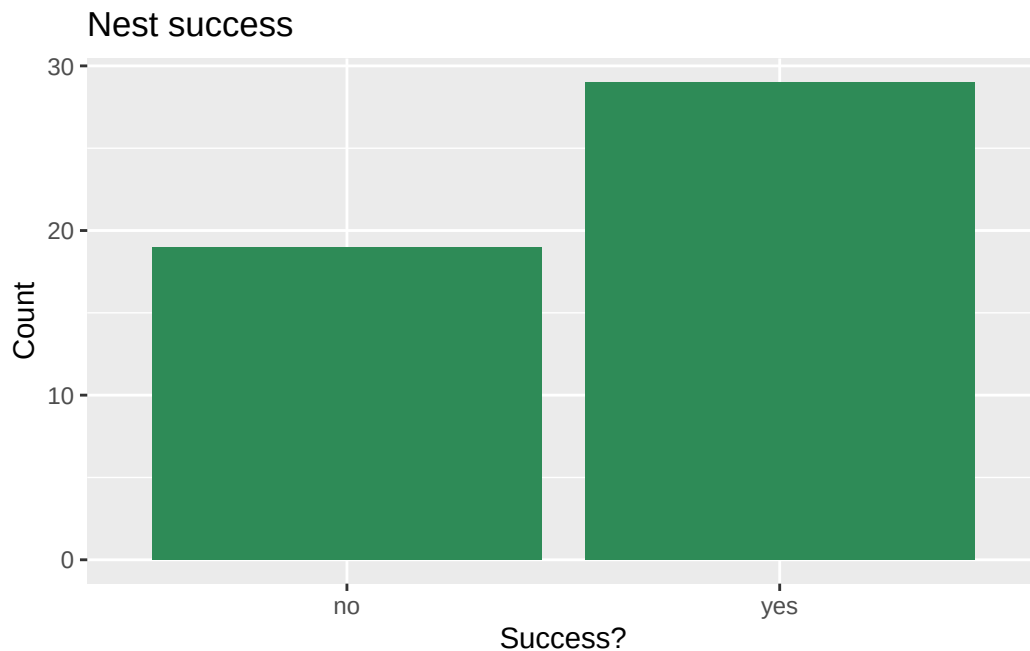


Figure 3: Binary outcome: nest success yes/no.

Interpret: With $n = 48$ nests, if $x = 30$ successes, state $\hat{p} = x/n$ and the percentage.

Answer: $\hat{p} = 30/48 = 0.625$, i.e. 62.5% of nests succeeded.

i Compute by hand (small datasets)

Work these before checking R. Same numbers appear in [Quiz A](#) and [Synthesis](#) practice banks.

Hand dataset 1 — Coffee orders

Order type	Count x
Latte	5
Drip	4
Tea	3

1. Total $n = 12$. Count for Latte: $x = 5$.
2. Proportion Latte: $\hat{p} = 5/12 \approx 0.417$ (about 41.7%).
3. Proportion not Latte: $7/12 \approx 0.583$.

Hand dataset 2 — Survey responses

Response	Count
Yes	13
No	7

$$\hat{p}_{\text{Yes}} = 13/20 = 0.65 \text{ (65\%).}$$

Hand dataset 3 — Bird feeder visits

Species	Count
Robin	12
Sparrow	11
Other	7

Robin proportion: $12/30 = 0.40$. Sparrow: $11/30 \approx 0.367$.

i From this sample to a population — inference reference note

Described here: the sample proportion $\hat{p} = x/n$ in a category. The corresponding **Population parameter** is p , the true proportion of the *whole population* in that category.

If all you have is a random sample but you want to **estimate** p with a confidence interval or **test a population claim** $H_0 : p = p_0$, then — after checking conditions (random sample; roughly at least 10 in and 10 out of the category) — use:

```
prop.test(x, n)      # approximate CI (lower, upper) and test for one proportion
binom.test(x, n)     # exact version for small counts
```

These were covered in **Module 1** (testing one proportion) and **Module 2** (estimating a proportion).

Check yourself — One categorical variable

1. Classify: (a) island name, (b) body mass in grams, (c) sex male/female.
2. Of $n = 40$ birds, $x = 10$ are on Dream island. Find $\hat{p}$ and express as a percentage.
3. Which graph is standard for one categorical variable: histogram, bar chart, or scatterplot?
4. Which tidyverse function counts rows per level of a categorical variable?

Answers — One categorical variable

1. (a) Categorical. (b) Numerical. (c) Categorical, binary.
2. $\hat{p} = 10/40 = 0.25$ — 25% of birds are on Dream island.
3. Bar chart (one bar per category).
4. `count(variable)` — returns one row per level with column `n`.

Section 2: One numerical variable

i Supplemental videos and reading for histograms and boxplots

Histograms

Resource	Type
Making a histogram — Khan Academy	Video
Interpreting histograms — Khan Academy	Video
Histograms — OpenStax Statistics	Reading

Boxplots

Resource	Type
Constructing a box plot — Khan Academy	Video
Reading box plots — Khan Academy	Video
Interpreting box plots — Khan Academy	Video
Box plots — OpenStax Statistics	Reading

Describing one numerical distribution: center, spread, shape, unusual values

What do people look for when they look at a histogram or dot plot? Usually we are interested in four features:

Feature	What to look for
Center	Where do values cluster?
Spread	How far apart are values?
Shape	Symmetric, skewed left, skewed right, or multi-modal?
Unusual observations	Outliers or gaps far from the main bulk

Center: mean and median

A numerical variable takes values that are counts or measurements where it is meaningful to sort them in order and to compute distance between any two of them. We describe such a variable by its **center**, **spread**, **shape**, and **unusual values**. We start with center — the two everyday summaries are the mean and the median.

Mean (average). For values $x_1, \dots, x_n$, the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The capital Greek letter Sigma (Σ) denotes summation. The expression $\sum_{i=1}^n x_i$ means: add $x_1 + x_2 + \dots + x_n$, including every observation of x , indexed by i from 1 to n . Multiplying that total by $\frac{1}{n}$ rescales it: you are taking the combined amount (the total sum) and expressing it as if it belonged to a single unit, “per person,” “per observation,”

or “per employee” in a payroll setting.

How to read the mean. Think: total of all values, spread evenly across n slots. If everyone’s contribution were replaced by the same number while keeping the same total, that common number would be the mean.

Visual (balance point). On a dot plot with one dot per observation, the mean is the balance point (center of mass) along the horizontal axis: equal weight at each dot, and the axis would balance at $\bar{x}$. On a histogram, you get the same idea if you treat each bin’s count (or properly scaled density) as sitting at the bin center, that is a standard approximation; the exact mean always comes from the original values, not from the drawn bins.

Median. Sort the values from smallest to largest. If n is odd, the median is the middle value. If n is even, it is the average of the two middle values. In the usual case with no ties sitting exactly on the split, the median is the value such that half of the observations are below it and half are above it.

How to interpret the median. The median is a positional measure: it marks the center of the ordered list of the observed values, not the balance of the total of the values.

Example: salaries. Suppose you list every employee’s annual salary. The median answers a “typical employee” question: half of the staff earn less than or equal to that amount and half earn more than or equal to it. The median is a useful picture of how pay feels in the middle of the workforce. The mean answers a “budget” question: if the firm paid the same total in salaries but split it evenly across every employee, each would get the mean. So the mean is the per-employee share of the fixed payroll total; it is what you need if you want to think about resizing the organization (“what if we had twice as many people with the same total wage bill?”) or track aggregate cost per head.

Which should you report? The choice is about purpose, not about shape or outliers. Decide what the number is *for*:

- A typical member’s experience $\rightarrow$ median. The median answers “what is it like for someone in the middle?” Half the values fall at or below it and half at or above, so it describes the standing of a typical unit no matter how large or small the extremes are.
- A total shared equally across units $\rightarrow$ mean. The mean is the total divided by n : the per-unit share you would get if the same total were split evenly. Reach for it whenever the *sum* is the thing that matters — budgets, rates, or scaling the group up or down.

Questions where the median fits:

- Home prices in a town. A buyer asking “what does a typical house cost?” wants the middle of the market — the number a realtor quotes — not a figure inflated by a few mansions.
- Household income. “How is a typical family doing?” is a question about the middle household, so the median income answers it.
- Clinic wait time. “How long will I probably wait?” is the experience of the middle patient — the median wait.

Questions where the mean fits:

- Payroll planning. “If we keep the same total wage bill, what does each employee cost on average?” is, by definition, total payroll $\div$ head count — the mean salary.
- Fuel for a fleet. “How much fuel should we budget per vehicle?” is total fuel $\div$ number of vehicles — again a mean, because the total is what must be covered.
- Buying tacos for lunch. Suppose I ask 10 students how many tacos each could eat, to estimate how many to buy for the whole class. A few big eaters make the data skewed, but I still want the mean: ordering enough tacos is about the *total*, and total tacos needed $\approx$ mean per student $\times$ number of students. Here the mean is the right answer **even though the distribution is skewed** — because the question is about a total, the skew does not change the choice.

Mean and median: examples to try by hand

1. Data: 7, 8, 9, 10, 10, 11, 12, 13 .

- Mean: $\bar{x} = (7 + 8 + 9 + 10 + 10 + 11 + 12 + 13)/8 = 80/8 = 10$.
- Median: notice that the 8 values are already sorted from smallest to biggest and the two middle values are both 10,

so the median is 10.

Here the mean and median match.

2. Data: 2, 3, 4, 5, 6, 7, 100 (seven values). One value (100) is far from the rest.

- Median: with seven values already sorted, the median is the fourth in the list so the median is 5.
- Mean: the sum is $2 + 3 + 4 + 5 + 6 + 7 + 100 = 127$, so $\bar{x} = 127/7 \approx 18.1$.

The mean is pulled toward the large observation; the median stays with the middle of the bulk of the data. For describing a “typical” case here, the median (5) is the point at which about half the values are less than it and half are greater than it. The mean reflects the total (127) spread across seven slots, including the one very large contribution and is the “balance point” of all seven values.

Mean and median in the tidyverse (pipe)

Syntax note: In dplyr, `summarize()` and `summarise()` are the same function (American vs. British spelling). For pipes, base R’s native `|>` (R 4.1+) works like the magrittr pipe `%>%` from the tidyverse: both pass the left-hand side into the first argument of the function on the right. This lecture uses `|>` and `summarize()`; use whichever spelling and pipe you prefer, they behave the same here.

The pipe `|>` sends the data on its left into the first argument of the function on its right. We often pipe a data frame (e.g. a dataset with examples in rows and variables in columns) into `summarize()` to compute summaries.

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {  
  install.packages("tidyverse", repos = "https://cloud.r-project.org")  
}
```

```
library(tidyverse)
```

```
scores <- tibble(  
  score = c(7, 8, 9, 10, 10, 11, 12, 13)  
)
```

```
scores |>  
  summarize(  
    n = n(),  
    mean_score = mean(score),  
    median_score = median(score)  
  )
```

```
# A tibble: 1 x 3  
      n mean_score median_score  
  <int>      <dbl>      <dbl>  
1     8         10         10
```

Reading the code: three kinds of words

That short chunk is worth pausing on, because every line of tidyverse code is built from three different kinds of words, and telling them apart is a super useful skill for reading (and later writing) R.

1. **Names a person chose.** Someone made up these labels; they are not special to R.

- `scores` is the name of the dataset (a **tibble** in the **tidyverse**). Whoever created the data picked that name, it could just as well have been `my_data` or `quiz`.
- `n`, `mean_score`, and `median_score` (the words on the **left** of each `=`) are names *you* choose for the output columns. You could call them `count`, `avg`, and `mid` and the code would work exactly the same.

2. **Words that come from the dataset.** These must match what is actually in the data, spelled exactly.
 - `score` (on the **right**, inside `mean(score)` and `median(score)`) is an existing **column** inside `scores`. You cannot invent it, if the column is really named `Score` or `points`, you must use that.
3. **R functions and operators.** These names are fixed by R or by a package; you cannot rename them.
 - `summarize()` collapses many rows into a one-row summary; `n()` counts the rows (one row per student here); `mean()` and `median()` compute the average and the middle value. `|>` is the **pipe**: it takes the thing on its left and feeds it as the first input to the function on its right. Read it as the word “then” — *take `scores`, then summarize it.*

Token in the code	What it is	Who decides the name?
<code>scores</code>	the dataset (a tibble) created earlier	a name you choose
<code>score</code>	a column inside scores	fixed by the data — must match exactly
<code>n =, mean_score =, median_score =</code> (left of =)	labels for the output columns	names you choose
<code>summarize, n, mean, median,</code> and the pipe	R functions / operators	fixed by R / the package

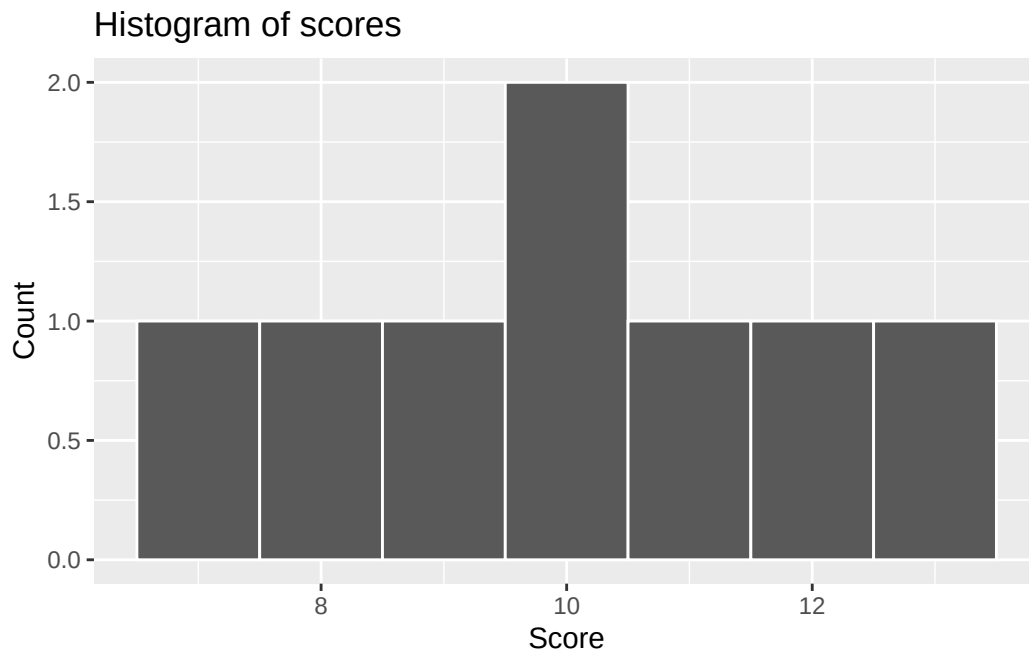
A quick test when you are unsure: *if you could rename it throughout and the code still ran, it is a name you chose.* Renaming `scores` to `my_data` everywhere is fine; renaming the column `score` (without changing the data) or the function `mean` would break the code. The same three buckets appear in every chunk in this lecture: dataset names like `penguins` and `mpg` and column names like `species` or `flipper_length_mm` come from the data; `count()`, `mutate()`, `group_by()`, and `|>` are R functions/operators; and anything you write on the left of an `=` is a label you picked.

Histogram and dot plot (ggplot2)

Histogram, split the axis into bins and show counts (or densities) in each bin.

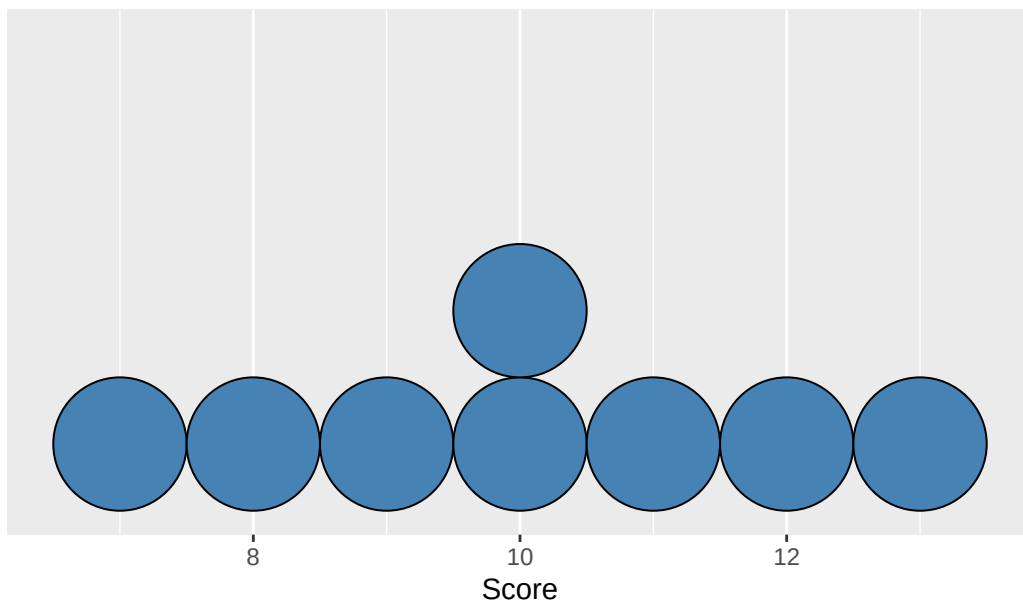
Dot plot, stack dots along the axis so you see each value (or rounded value) and repeats clearly.

```
ggplot(scores, aes(x = score)) +
  geom_histogram(binwidth = 1, boundary = 6.5, color = "white") +
  labs(
    title = "Histogram of scores",
    x = "Score",
    y = "Count"
  )
```



```
ggplot(scores, aes(x = score)) +
  geom_dotplot(binwidth = 1, method = "histodot", fill = "steelblue") +
  labs(
    title = "Dot plot of scores",
    x = "Score",
    y = NULL
  ) +
  scale_y_continuous(NULL, breaks = NULL) # hide stacking axis
```

Dot plot of scores



Adjust `binwidth` so the plot matches how you think about the scale (e.g. group by every 1 unit, 10 units, 100 units...) or leave this input out to let R use a default setting.

Spread: why the center is not the whole story

The mean and median answer one question: “what is a typical value?” But the center alone can hide a lot. Two datasets can share the *same* center and still describe very different situations: in one, the values huddle tightly around the center; in the other, they scatter far above and below it. If you reported only the mean, those two datasets would look identical on paper.

So we add a second idea: **spread**, which is how far apart the values are. The summaries in this section (range, standard deviation, IQR, and the absolute-deviation measures) each put a number on that distance, the same way the mean and median put a number on the center. The two datasets below are built to make the point: they have an *identical* mean, yet their spreads are very different.

Two small datasets to try by hand

Suppose we have two set of numbers, eight values per group:

Dataset N (“narrow”): 7, 8, 9, 10, 10, 11, 12, 13

Dataset W (“wide”): 4, 6, 8, 10, 10, 12, 14, 16

Both have mean $\bar{x} = 10$ (check: sums are 80 each). The spread differs: N is clustered near 10; W stretches toward 4 and 16.

Range

- N: $\max - \min = 13 - 7 = 6$.
- W: $16 - 4 = 12$.

Standard deviation (sample)

First, with no sigma notation: for n observations $x_1, x_2, \dots, x_n$ with mean $\bar{x}$, the sample standard deviation is

$$s = \sqrt{\frac{1}{n-1} [(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2]}.$$

You average the squared deviations from the mean (add them all, then divide by $n - 1$), then take the square root so the result is on the same scale as the data.

The same quantity in sigma (summation) notation is

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}.$$

Here $\sum_{i=1}^n (x_i - \bar{x})^2$ means “add $(x_i - \bar{x})^2$ for $i = 1, 2, \dots, n$,” i.e. the numerator inside the square root in the expanded form above.

By hand for dataset N ($n = 8$, $\bar{x} = 10$). Put the data in column 1, then add columns for the deviation from the mean and its square. Total column 3 to get the sum of squared deviations (often called SS).

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
7	-3	9
8	-2	4
9	-1	1
10	0	0
10	0	0
11	1	1
12	2	4

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
13	3	9

Sum of column 3: $9 + 4 + 1 + 0 + 0 + 1 + 4 + 9 = 28$.

The sample variance is the average of the squared deviations using divisor $n - 1$:

$$s^2 = \frac{28}{8-1} = \frac{28}{7} = 4.$$

(Aside: s^2 is the variance; it has nicer algebra but is in squared units.) The sample SD is the square root:

$$s = \sqrt{4} = 2.$$

Dataset W ($\bar{x} = 10$). Repeat the same steps: column 1 is x , column 2 is $x - \bar{x}$, column 3 is $(x - \bar{x})^2$, then sum column 3 and finish with variance and SD. Use the space below in the PDF.

Answer (dataset W).

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
4	-6	36
6	-4	16
8	-2	4
10	0	0
10	0	0
12	2	4
14	4	16
16	6	36

Sum of column 3: $36 + 16 + 4 + 0 + 0 + 4 + 16 + 36 = 112$.

$$s^2 = \frac{112}{7} = 16, \quad s = \sqrt{16} = 4.$$

Same mean, different SD, W is twice as variable as N in this sense.

Interpreting SD

A single SD is hard to interpret on its own. It is measured in the same units as the data, but “2” or “4” means little without context, and it does not correspond to a distance you can point to in the data — unless the distribution is close to normal. In that normal case the empirical rule gives it meaning: roughly 68% of values fall within one SD of the mean, 95% within two SDs, and 99.7% within three SDs. Real one-variable data are rarely exactly normal, so for a single variable that reading is approximate at best.

What SD is really for. Like the choice between mean and median, the choice of spread measure is about purpose. The standard deviation’s main job is as a *building block for inference*, not a stand-alone description: it feeds the standard error $s/\sqrt{n}$, and therefore the confidence intervals and t -tests from Module 2 (and later in this module). It is also the natural scale for comparing the spread of the same outcome across groups or studies (as with N vs. W above). Reach for SD when you are headed toward a CI, a test, or a model — or when you specifically want the mean’s companion measure.

IQR and MAD for describing one variable. When the goal is simply to describe the spread of one variable, the IQR and the MAD usually read more naturally. The IQR is the width of the middle 50% of the data — a span you can mark directly on the number line. The MAD is a typical distance from the median. Both are in the data’s own units, both pair with the median, and neither needs a normality assumption to mean something — they are the spread analogues of the median, describing the typical experience of the data rather than feeding a downstream calculation.

Optional: mean and median absolute deviation from the median

Two related summaries use absolute (not squared) distances from the median $\tilde{x}$. They are easy to interpret for one group or for comparing groups.

Mean absolute deviation from the median (average distance to the median):

$$\frac{1}{n} \sum_{i=1}^n |x_i - \tilde{x}|.$$

Median absolute deviation (often written MAD): take the median of the n numbers $|x_1 - \tilde{x}|, \dots, |x_n - \tilde{x}|$. Half of those absolute deviations are at or below MAD and half are at or above, so MAD is a typical magnitude of distance from the median (robust to outliers in the same spirit as the median itself).

For dataset N, $\tilde{x} = 10$. The absolute deviations $|x_i - 10|$ are 3, 2, 1, 0, 0, 1, 2, 3.

- Mean absolute deviation from the median: $(3 + 2 + 1 + 0 + 0 + 1 + 2 + 3)/8 = 12/8 = 1.5$.
- Median absolute deviation: sorted deviations 0, 0, 1, 1, 2, 2, 3, 3, so $\text{MAD} = (1 + 2)/2 = 1.5$.

For dataset W, $\tilde{x} = 10$. The values $|x_i - 10|$ are 6, 4, 2, 0, 0, 2, 4, 6.

- Mean absolute deviation from the median: $(6 + 4 + 2 + 0 + 0 + 2 + 4 + 6)/8 = 24/8 = 3$.
- Median absolute deviation: sorted 0, 0, 2, 2, 4, 4, 6, 6, so $\text{MAD} = (2 + 4)/2 = 3$.

Interpretation. The mean absolute deviation from the median answers: on average, how far are observations from the median? Here, about 1.5 units (N) vs 3 units (W), matching the idea that W is more spread out. The MAD gives a typical absolute distance from the median (half of the absolute deviations fall at or below it, half at or above); for N and W it matches the mean absolute deviation here because the deviation lists are symmetric. Either measure is useful alone or side by side with SD when you compare groups.

Quartiles and IQR by Hand

Sort the data. For eight values, the median is the average of the 4th and 5th values. Split into lower four and upper four; Q_1 is the median of the lower half, Q_3 the median of the upper half.

N: sorted 7, 8, 9, 10, 10, 11, 12, 13. Median = $(10 + 10)/2 = 10$. Lower half 7, 8, 9, 10 $\Rightarrow Q_1 = (8 + 9)/2 = 8.5$. Upper half 10, 11, 12, 13 $\Rightarrow Q_3 = (11 + 12)/2 = 11.5$. IQR = $11.5 - 8.5 = 3$.

W: sorted 4, 6, 8, 10, 10, 12, 14, 16. Median = 10. Lower half 4, 6, 8, 10 $\Rightarrow Q_1 = 7$. Upper half 10, 12, 14, 16 $\Rightarrow Q_3 = 13$. IQR = 6.

Second example (odd n , outliers). Take nine values so the median is a single order statistic (the middle of the sorted list). Use the same rule as above: sort the data, take the median, then form the lower and upper halves excluding the median (here each half has four values), and take the median of each half as Q_1 and Q_3 . For four numbers, the median is the average of the two middle values in that half.

Dataset A (symmetric): 5, 6, 7, 8, 9, 10, 11, 12, 13. Sorted: 5, 6, 7, 8, 9, 10, 11, 12, 13.

- $n = 9$ is odd, so the median is the 5th value: $\tilde{x} = 9$.
- Lower half (four values below the median): 5, 6, 7, 8 $\rightarrow Q_1 = (6 + 7)/2 = 6.5$.
- Upper half (four values above the median): 10, 11, 12, 13 $\rightarrow Q_3 = (11 + 12)/2 = 11.5$.
- $IQR = 11.5 - 6.5 = 5$.

Dataset B (two large values in the upper tail): 5, 6, 7, 8, 9, 10, 11, 90, 100. Sorted: 5, 6, 7, 8, 9, 10, 11, 90, 100.

- The median is still the 5th value: $\tilde{x} = 9$.
- Lower half: 5, 6, 7, 8 $\rightarrow Q_1 = 6.5$ (unchanged from A).
- Upper half: 10, 11, 90, 100 $\rightarrow Q_3 = (11 + 90)/2 = 50.5$ (the two middle values in the upper four are 11 and 90).
- $IQR = 50.5 - 6.5 = 44$.

So Q_3 and IQR jump when the upper four values include extremes that move the middle pair used for Q_3 . With $n = 9$, a single outlier at the very top often does not change Q_3 (the upper half still has four values, and the middle two can stay small), which is why this illustration uses two large upper values. By contrast, dataset N and W above use even $n = 8$, where the overall median is the average of the two middle order statistics and each half has four observations. Same Tukey-style splitting idea.

Five-number summary: the values of min, Q_1 , median, Q_3 , max break the dataset into 4 roughly equally sized pieces.

Note: Software may use slightly different quartile interpolation; match your course's convention and `?quantile` in R if you need exact agreement.

Shape: symmetry and skew

Now that we have a center (mean and median) and a measure of spread, we can describe **shape**: how the values are arranged around the center.

- Symmetric: the left and right halves are near mirror images — fold the histogram down the middle and the two sides roughly match. The mean and median are close together.
- Right-skewed (positively skewed): a long tail stretches toward the high (right-hand) values, while most observations bunch on the left. The few large values pull the mean above the median.
- Left-skewed (negatively skewed): a long tail stretches toward the low (left-hand) values, while most observations bunch on the right. The few small values pull the mean below the median.

The skew is named for the side the long tail points to — a tail trailing off to the right is right-skewed; a tail trailing off to the left is left-skewed. This is exactly why the mean–median comparison from the previous part is a quick skew check: when the mean sits noticeably above the median, suspect a right tail; below, a left tail.

The three histograms below show the same idea on simulated data. Watch where the long tail points and where the mean falls relative to the median (the dashed and solid lines).

```
library(tidyverse)
set.seed(311)
sym <- tibble(x = rnorm(1000, mean = 50, sd = 8))

ggplot(sym, aes(x = x)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  geom_vline(aes(xintercept = median(x)), linewidth = 1) +
  geom_vline(aes(xintercept = mean(x)), linetype = "dashed", linewidth = 1) +
```

```
labs(title = "Symmetric", x = "Value", y = "Count")
```

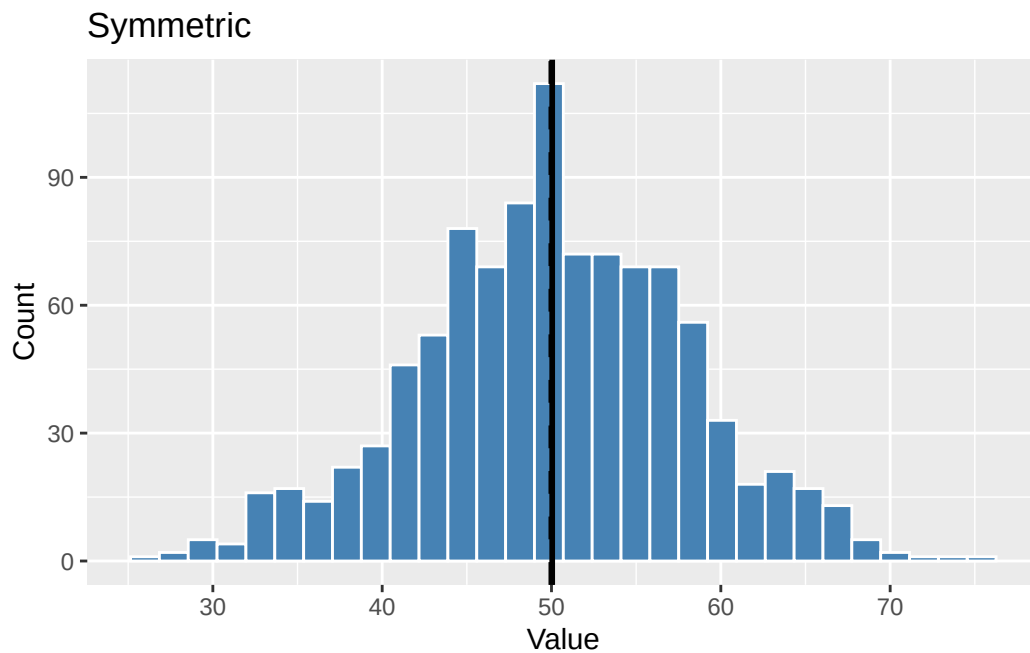


Figure 4: Symmetric: the two halves are near mirror images, so the mean (dashed) and median (solid) nearly coincide.

```
set.seed(312)
right <- tibble(x = rexp(1000, rate = 1 / 10))

ggplot(right, aes(x = x)) +
  geom_histogram(bins = 30, fill = "darkorange", color = "white") +
  geom_vline(aes(xintercept = median(x)), linewidth = 1) +
  geom_vline(aes(xintercept = mean(x)), linetype = "dashed", linewidth = 1) +
  labs(title = "Right-skewed (positive skew)", x = "Value", y = "Count")
```

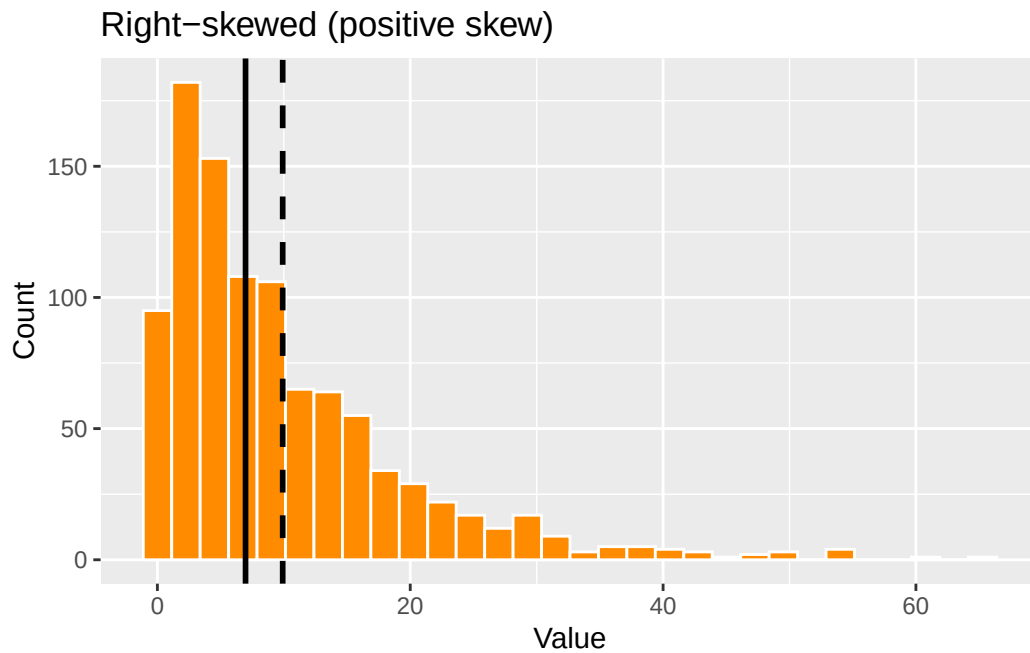


Figure 5: Right-skewed: the long tail points right toward high values, pulling the mean (dashed) above the median (solid).

```
set.seed(313)
left <- tibble(x = 100 - rexp(1000, rate = 1 / 10))

ggplot(left, aes(x = x)) +
  geom_histogram(bins = 30, fill = "seagreen", color = "white") +
  geom_vline(aes(xintercept = median(x)), linewidth = 1) +
  geom_vline(aes(xintercept = mean(x)), linetype = "dashed", linewidth = 1) +
  labs(title = "Left-skewed (negative skew)", x = "Value", y = "Count")
```

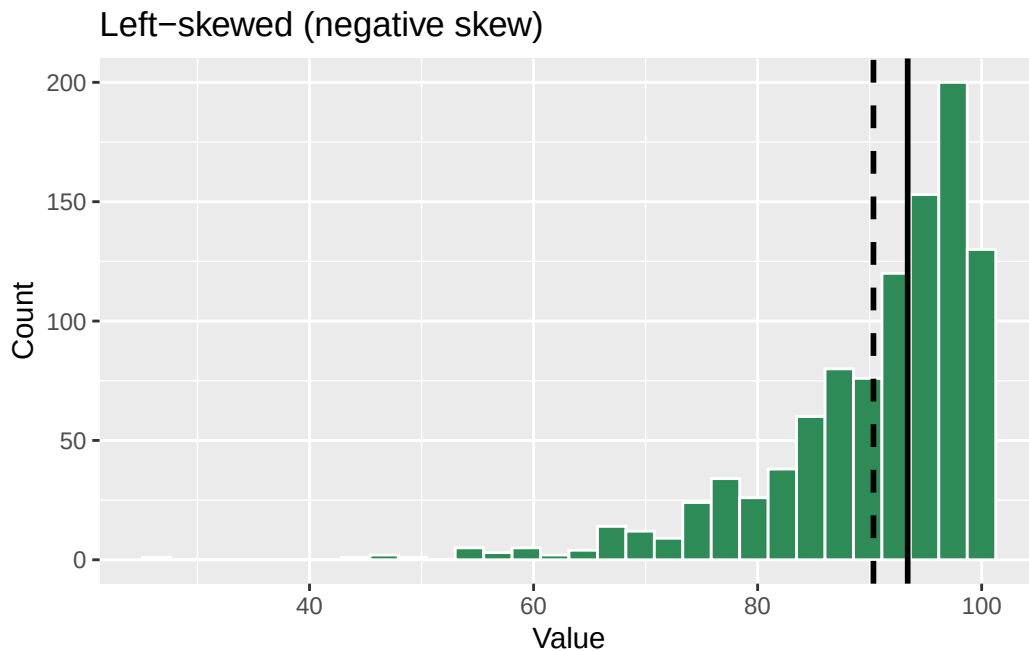


Figure 6: Left-skewed: the long tail points left toward low values, pulling the mean (dashed) below the median (solid).

Unusual observations and outliers

An **outlier** is a value that sits far from the bulk of the data. A common rule of thumb uses the IQR to draw “fences”:

$$\text{lower fence} = Q_1 - 1.5 \times \text{IQR}, \quad \text{upper fence} = Q_3 + 1.5 \times \text{IQR}.$$

Any value below the lower fence or above the upper fence is flagged as a potential outlier. This is exactly the rule a boxplot uses to decide which points to draw separately beyond the whiskers. The fences scale with spread (IQR), so a tightly clustered variable flags points sooner than a widely spread one. A flagged point is not automatically an error (it may be a genuine extreme value) but it is worth a second look.

Boxplots and the five-number summary

A boxplot shows the five-number summary: the box runs from Q_1 to Q_3 (spanning the middle 50%), the median is drawn inside, and whiskers extend to sensible extremes (often the furthest points within $1.5 \times \text{IQR}$ of the quartiles, with more extreme points as outliers). Rules differ by implementation, check `?geom_boxplot`.

For N and W, sketch two boxes side by side: same median (10), but W has a wider box and whiskers, matching larger IQR and range.

💡 Interpret the output — code comes in the demo lab

The chunks below use tidyverse to produce tables and figures. Focus on reading summaries and graphs. You will learn how to write these steps in [Demo 1](#) and [Demo 2](#).

Example 1 — Flipper length (mm)

```
library(tidyverse)
if (!requireNamespace("palmerpenguins", quietly = TRUE)) {
  install.packages("palmerpenguins", repos = "https://cloud.r-project.org")
}
library(palmerpenguins)

penguins |>
  summarize(
    n = sum(!is.na(flipper_length_mm)),
    mean_mm = mean(flipper_length_mm, na.rm = TRUE),
    median_mm = median(flipper_length_mm, na.rm = TRUE),
    sd_mm = sd(flipper_length_mm, na.rm = TRUE),
    iqr_mm = IQR(flipper_length_mm, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 5
      n mean_mm median_mm sd_mm iqr_mm
<int>   <dbl>     <dbl> <dbl> <dbl>
1   342    201.      197  14.1    23
```

```
ggplot(penguins, aes(x = flipper_length_mm)) +
  geom_histogram(binwidth = 5, boundary = 180, fill = "steelblue", color = "white") +
  labs(title = "Flipper length", x = "Flipper length (mm)", y = "Count")
```

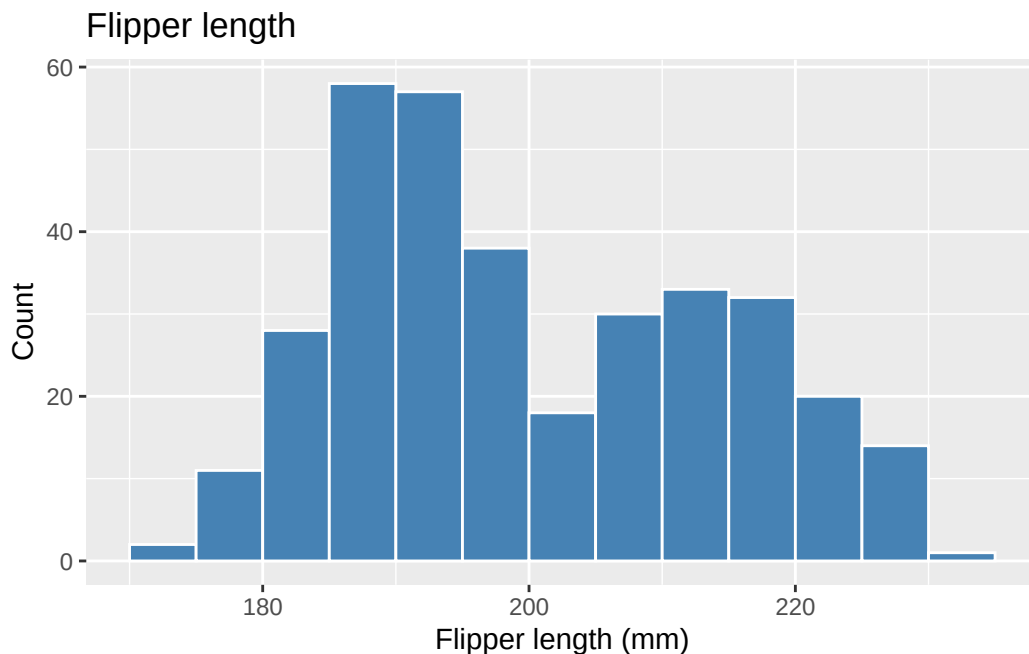


Figure 7: Distribution of flipper length — roughly symmetric, single peak.

Interpret: Center (mean vs median), spread (SD or IQR), shape (symmetric/skewed?), any unusual values?

Answer: Center — mean ≈ 200.9 mm and median = 197 mm are close. Spread — SD ≈ 14.1 mm (IQR ≈ 23 mm).

Shape — roughly symmetric with a single main peak; no extreme outliers.

Example 2 — Body mass (g) — right-skew common in biology

```
penguins |>
  summarize(
    mean_g = mean(body_mass_g, na.rm = TRUE),
    median_g = median(body_mass_g, na.rm = TRUE),
    sd_g = sd(body_mass_g, na.rm = TRUE),
    iqr_g = IQR(body_mass_g, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 4
  mean_g median_g sd_g iqr_g
  <dbl>   <dbl> <dbl> <dbl>
1  4202.   4050  802. 1200
```

```
ggplot(penguins, aes(x = body_mass_g)) +
  geom_histogram(bins = 20, fill = "chocolate4", color = "white") +
  labs(title = "Body mass", x = "Body mass (g)", y = "Count")
```

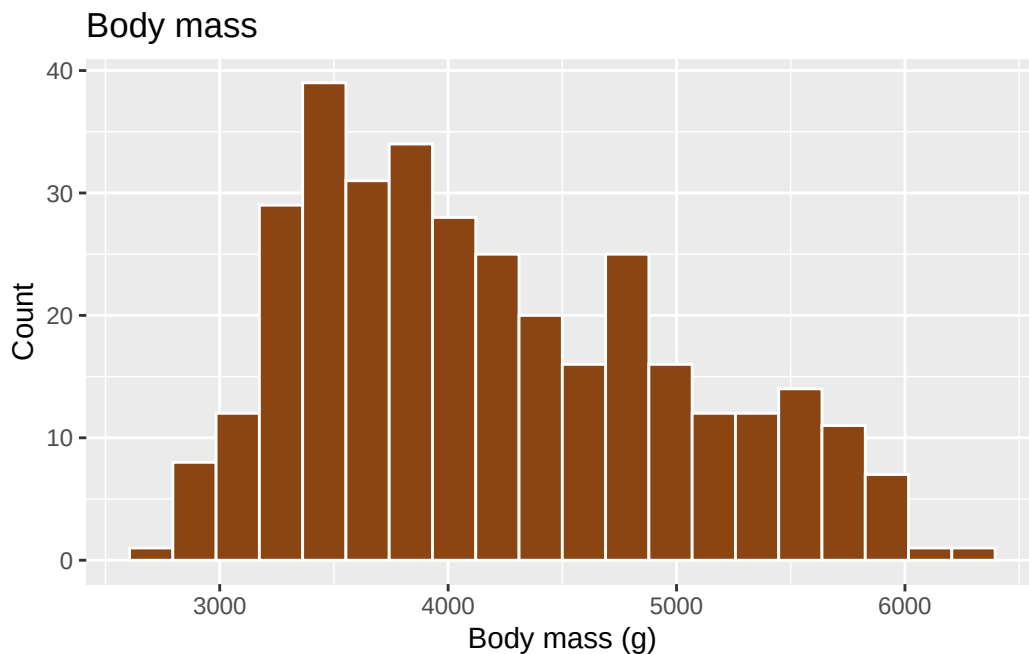


Figure 8: Body mass distribution — often right-skewed when a few heavy individuals exist.

Interpret: Is the mean above the median? That pattern suggests right skew or heavy upper tail.

Answer: Yes — mean $\approx$ 4202 g vs median = 4050 g, so the distribution is mildly right-skewed (a heavier upper tail).

Example 3 — Highway MPG (mpg)

```
data("mpg", package = "ggplot2")
mpg |>
```

```
summarize(
  mean_hwy = mean(hwy, na.rm = TRUE),
  median_hwy = median(hwy, na.rm = TRUE),
  sd_hwy = sd(hwy, na.rm = TRUE),
  range_hwy = max(hwy) - min(hwy)
)
```

```
# A tibble: 1 x 4
  mean_hwy median_hwy sd_hwy range_hwy
  <dbl>      <dbl> <dbl>    <int>
1    23.4         24  5.95     32
```

```
ggplot(mpg, aes(x = hwy)) +
  geom_dotplot(binwidth = 1, method = "histodot", fill = "olivedrab") +
  labs(title = "Highway MPG", x = "Highway MPG", y = NULL) +
  scale_y_continuous(NULL, breaks = NULL)
```

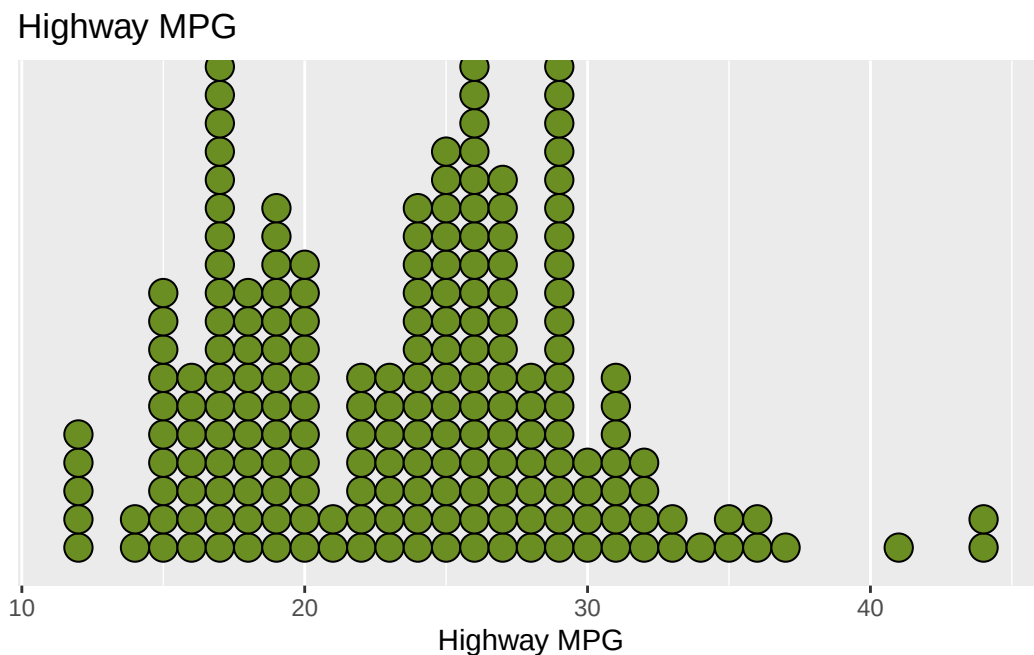


Figure 9: Dot-style view of highway MPG (each dot is one car model).

Interpret: Report the range and one measure of center. Which center fits your question, a typical model's mileage (median, paired with IQR or MAD) or an average tied to a total, such as fuel budgeted across many cars (mean, paired with SD)?

Answer: Range = $44 - 12 = 32$ mpg. A typical model's mileage $\rightarrow$ median = 24 mpg; a fuel-budget-across-cars question $\rightarrow$ mean ≈ 23.4 mpg.

i Compute by hand (small datasets)

Work these before checking R. Same numbers appear in [Quiz A](#) and [Synthesis](#) practice banks.

Hand dataset 1 — Quiz scores ($n = 8$)

Data: 72, 74, 76, 78, 80, 82, 84, 90.

Statistic	By hand
Mean $\bar{x}$	$(72 + \dots + 90)/8 = 79.5$
Median	Average of 4th and 5th: $(78 + 80)/2 = 79$
Range	$90 - 72 = 18$
Q_1	Lower half median: $(74 + 76)/2 = 75$
Q_3	Upper half median: $(82 + 84)/2 = 83$
IQR	$83 - 75 = 8$

Hand dataset 2 — Daily steps ($n = 7$)

Data: 4,200; 5,100; 5,800; 6,000; 6,200; 6,500; 12,000.

- Mean $\approx 6,543$ (pulled up by 12,000).
- Median = 6,000 (middle value)
- Range = $12,000 - 4,200 = 7,800$.

Hand dataset 3 — Small sample SD check ($n = 5$)

Data: 10, 12, 14, 16, 18.

- Mean $\bar{x} = 14$.
- Deviations: $-4, -2, 0, 2, 4$; squared: $16, 4, 0, 4, 16$; sum = 40.
- Sample variance $s^2 = 40/(5 - 1) = 10$; $s = \sqrt{10} \approx 3.16$.

Optional: checking normality with a Q–Q plot

i Note: this section is not required for Project 3

This subsection is for curiosity and for later modules. Project 3 only asks you to describe shape in words (symmetric, skewed, outliers); you do not need Q–Q plots or formal normality checks.

Some methods (for example the empirical rule, and the one-sample t procedures from Module 2) work best when a numerical variable is roughly normal. A histogram gives a first impression, but a *normal quantile–quantile (Q–Q) plot* is a sharper visual check. It plots each sorted data value against the value you would expect if the data were exactly normal. If the points fall close to a straight reference line, the normal model is reasonable; systematic curves or S-shapes signal skew or heavy tails.

In ggplot2, map the variable to the `sample` aesthetic and add `stat_qq()` (the points) and `stat_qq_line()` (the reference line):

```
library(tidyverse)
if (!requireNamespace("palmerpenguins", quietly = TRUE)) {
  install.packages("palmerpenguins", repos = "https://cloud.r-project.org")
}
library(palmerpenguins)

ggplot(penguins, aes(sample = flipper_length_mm)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "Normal Q-Q plot: flipper length",
    x = "Theoretical normal quantiles",
    y = "Sample quantiles"
```


)

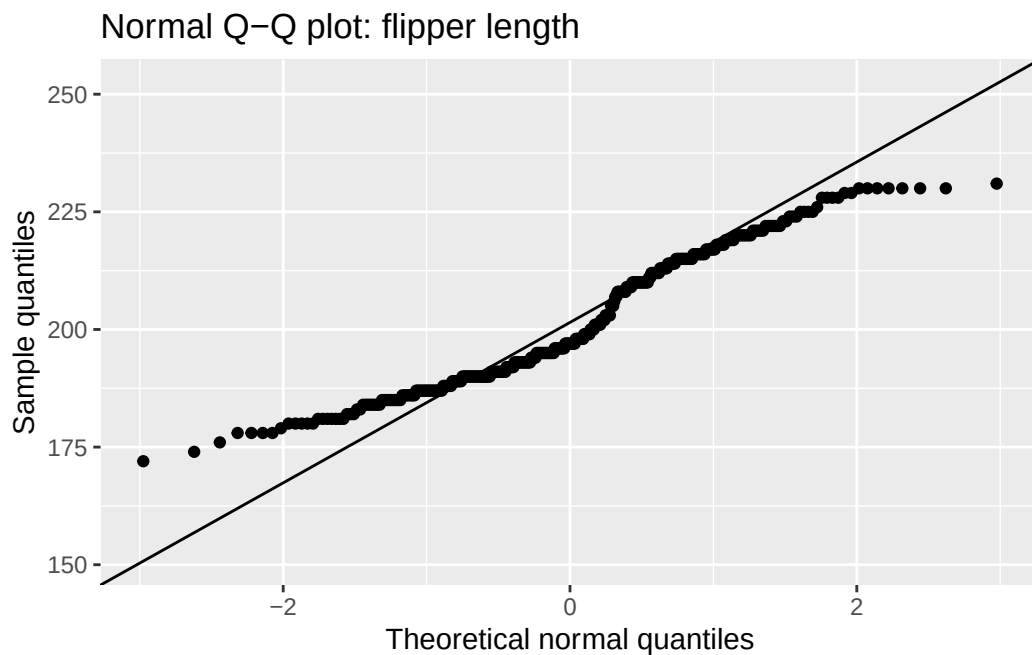


Figure 10: Q-Q plot of flipper length — points hug the line, so the normal model is reasonable.

Now compare a right-skewed variable: the points bend away from the line (curving up at the right), the Q-Q signature of a long upper tail.

```
set.seed(314)
skewed <- tibble(x = rexp(300, rate = 1 / 10))

ggplot(skewed, aes(sample = x)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "Normal Q-Q plot: right-skewed data",
    x = "Theoretical normal quantiles",
    y = "Sample quantiles"
  )
```

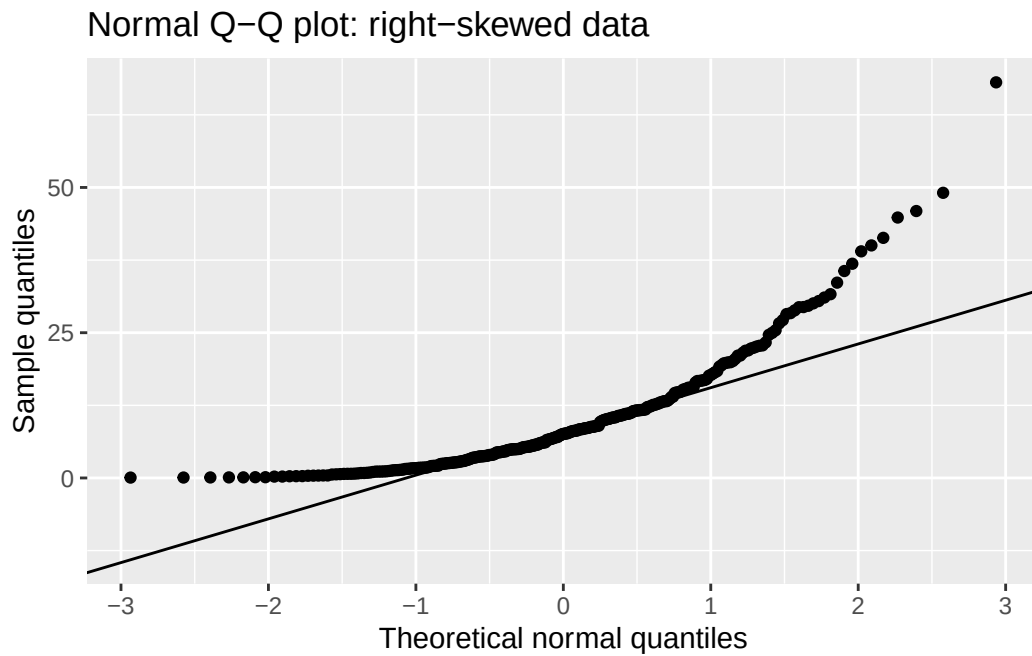


Figure 11: Q-Q plot of a right-skewed variable — points curve away from the straight line.

A left-skewed variable bends the opposite way: the points curve away from the line at the lower-left, the Q-Q signature of a long lower tail.

```
set.seed(315)
left_skewed <- tibble(x = 100 - rexp(300, rate = 1 / 10))

ggplot(left_skewed, aes(sample = x)) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "Normal Q-Q plot: left-skewed data",
    x = "Theoretical normal quantiles",
    y = "Sample quantiles"
  )
```

Normal Q–Q plot: left-skewed data

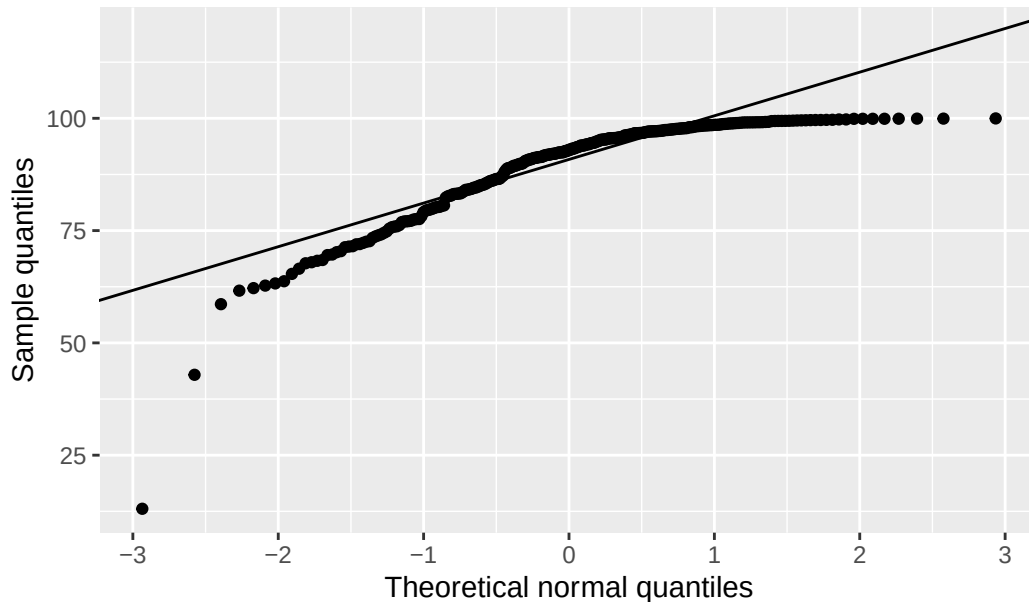


Figure 12: Q–Q plot of a left-skewed variable — points curve away from the line at the lower-left.

How to read it: a straight line means approximately normal; upward-curving ends mean right skew; downward-curving ends mean left skew; an S-shape means heavy or light tails. Use it together with the histogram, not instead of it.

i From this sample to a population — inference reference note

Described here: the sample mean $\bar{x}$ (and SD s). **Population parameter:** μ — the true mean of the *whole* population.

If all you have is a random sample but you want to **estimate** μ with a confidence interval or **test a claim** $H_0 : \mu = \mu_0$, then — after checking conditions (random sample; n large or the values roughly symmetric) — use the one-sample t -statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}},$$

whose denominator $s/\sqrt{n}$ is the standard error of the mean; under H_0 it is compared to a Student t distribution with $df = n - 1$. In practice you let software compute the p -value and interval:

```
t.test(x, mu = mu0)    # CI (lower, upper) and test for one mean
```

Covered in **Module 2** (confidence interval and test for one mean).

Check yourself — One numerical variable

1. Data: 2, 3, 4, 5, 6, 7, 100. Without calculating them which is larger — mean or median? Why?
2. Dataset N: 7, 8, 9, 10, 10, 11, 12, 13. Compute the IQR.
3. Name four features we describe when looking at one numerical distribution.
4. Which ggplot2 geometries show one numerical variable?
5. What five numbers define a boxplot?

Answers — One numerical variable

1. Mean (≈ 18.1) is pulled toward 100; median is 5 and better describes a typical value.
2. $Q_1 = 8.5$, $Q_3 = 11.5$. IQR = 3.
3. Center, spread, shape, unusual observations (outliers).
4. `geom_histogram()` and `geom_dotplot()`.
5. Minimum, Q_1 , median, Q_3 , maximum (five-number summary).

Section 3: One numerical and one categorical

i Bridge from Sections 1–2

Sections 1–2 treated one variable at a time. Here the numerical variable is measured separately within each level of a categorical variable (e.g. flipper length for each species). Compare center and spread per group with `group_by()` + `summarize()` and side-by-side boxplots (or grouped histograms / dot plots).

Setting: one numerical variable, two categories

We often compare a numerical outcome (e.g. recovery time, blood pressure) between two groups defined by a binary categorical variable (treatment vs. control, exposed vs. not exposed). Descriptively we compare center and spread; inferentially we may compare population means with a two-sample t -test.

A variable is not the same as its categories

This is one of the most common mix-ups, so it is worth slowing down. A **variable** is a single characteristic you measure — it becomes one **column** in your data. The **categories** (also called **levels** or **unique values**) of a categorical variable are the distinct labels that can appear in that column.

A **binary** categorical variable is **one variable with two categories** — not two variables. Saying “I have two categorical variables” because a column contains both **Drug** and **Placebo** is like saying a light switch is two switches because it can be either *on* or *off*. There is one switch (one variable); *on* and *off* are its two settings (two categories).

A helpful way to keep them straight:

Idea	Question it answers	Example
Variable	<i>What did we measure?</i>	treatment
Category (level)	<i>What answers are possible?</i>	Drug, Placebo
Value	<i>What did this one row record?</i>	Drug (for patient 1)

So the variable is the **question**, the categories are the **possible answers**, and each row holds one **value** (one of the possible answers).

Worked example: a 10-row, 2-column dataset

Suppose we record, for 10 patients, which arm of a trial they were in and how many days they took to recover:

treatment	recovery_days
Drug	5
Drug	6
Drug	4
Drug	7

treatment	recovery_days
Drug	5
Placebo	9
Placebo	11
Placebo	8
Placebo	10
Placebo	12

Now read this dataset carefully, one count at a time:

- **Columns = variables.** There are **2 columns**, so there are **2 variables**: `treatment` and `recovery_days`.
- **`treatment` is one categorical variable.** It has **2 categories** (levels): `Drug` and `Placebo`. Two categories does **not** mean two variables — it is still a single column.
- **`recovery_days` is one numerical variable.** Its values are counts of days, where order and distance matter.
- **Rows = observations (values), not categories.** The `treatment` column holds **10 values** (one per patient), but only **2 distinct values**. `Drug` appears 5 times and `Placebo` appears 5 times. The number of *categories* is the number of distinct labels (**2**), not the number of *rows* (10), and not the number of *variables* (2).

⚠ Common mistake

“This dataset has two categorical variables: `Drug` and `Placebo`.”

`Drug` and `Placebo` are **values**, not variables. They are the two **categories** of the single categorical variable `treatment`. The correct statement is: “one categorical variable (`treatment`, with two categories) and one numerical variable (`recovery_days`).”

This is exactly the “one numerical and one categorical variable” setting of this section: we compare the numerical variable (`recovery_days`) across the categories of the categorical variable (`treatment`). In notation, a categorical variable with k levels here has $k = 2$.

Check in R

Put both small datasets in one data frame: a numerical outcome `y` and a binary categorical `group` (Narrow vs Wide). Then use `group_by()` and `summarize()` for descriptives, and `ggplot2` for plots.

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

compare <- tibble(
  group = factor(
    rep(c("Narrow", "Wide"), each = 8),
    levels = c("Narrow", "Wide")
  ),
  y = c(
    7, 8, 9, 10, 10, 11, 12, 13,
    4, 6, 8, 10, 10, 12, 14, 16
  )
)
```

```
compare |>
  group_by(group) |>
  summarize(
    n = n(),
    mean_y = mean(y),
    sd_y = sd(y),
    median_y = median(y),
    min_y = min(y),
    max_y = max(y),
    range_y = max(y) - min(y),
    q1_R = quantile(y, 0.25, names = FALSE),
    q3_R = quantile(y, 0.75, names = FALSE),
    iqr = IQR(y),
    mad_from_median = median(abs(y - median(y))),
    mean_abs_dev_from_median = mean(abs(y - median(y)))
  )
```

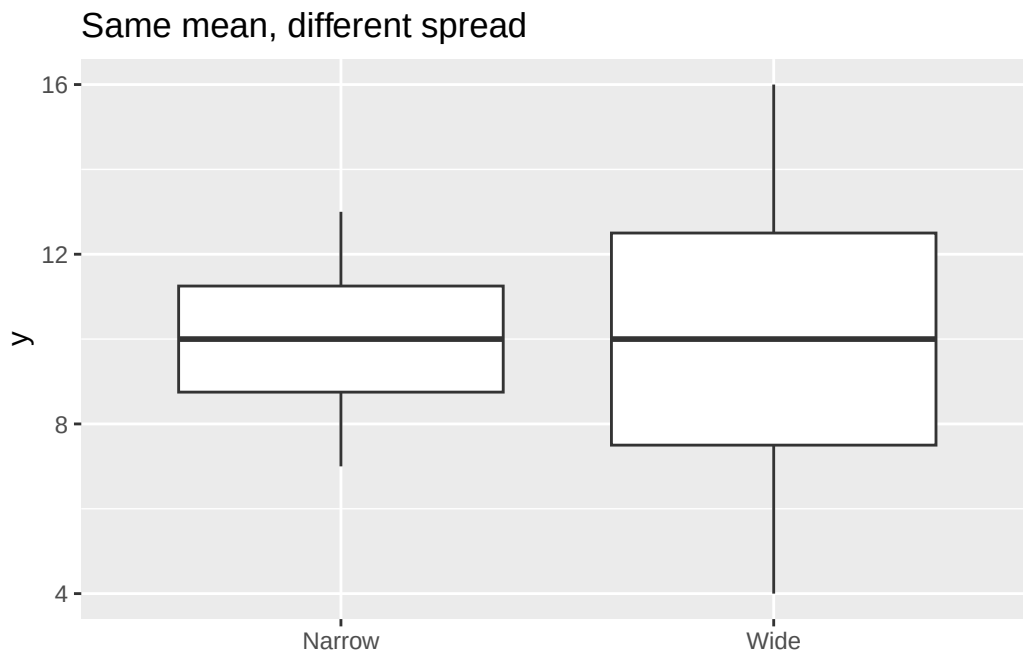
A tibble: 2 x 13

	group	n	mean_y	sd_y	median_y	min_y	max_y	range_y	q1_R	q3_R	iqr
	<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Narrow	8	10	2	10	7	13	6	8.75	11.2	2.5
2	Wide	8	10	4	10	4	16	12	7.5	12.5	5

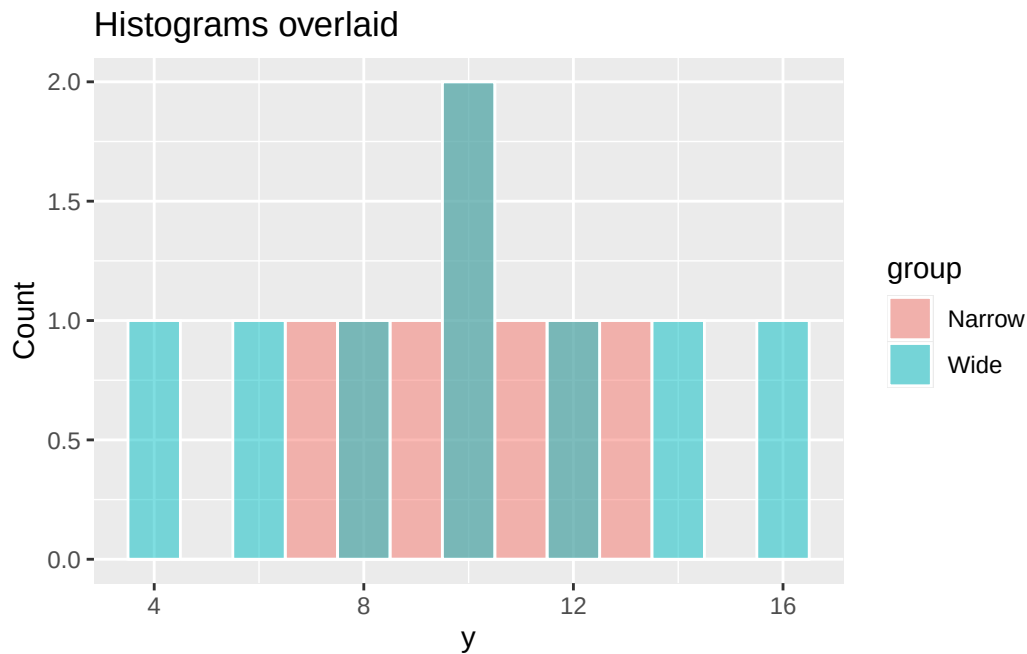
i 2 more variables: mad_from_median <dbl>, mean_abs_dev_from_median <dbl>

`quantile()` and `IQR()` use R's default definition (see `?quantile`), which may differ slightly from the Tukey-style hinges in the hand work above. MAD here is the median absolute deviation from the median; `mean_abs_dev_from_median` is the mean of those absolute deviations.

```
ggplot(compare, aes(x = group, y = y)) +
  geom_boxplot() +
  labs(title = "Same mean, different spread", x = NULL, y = "y")
```

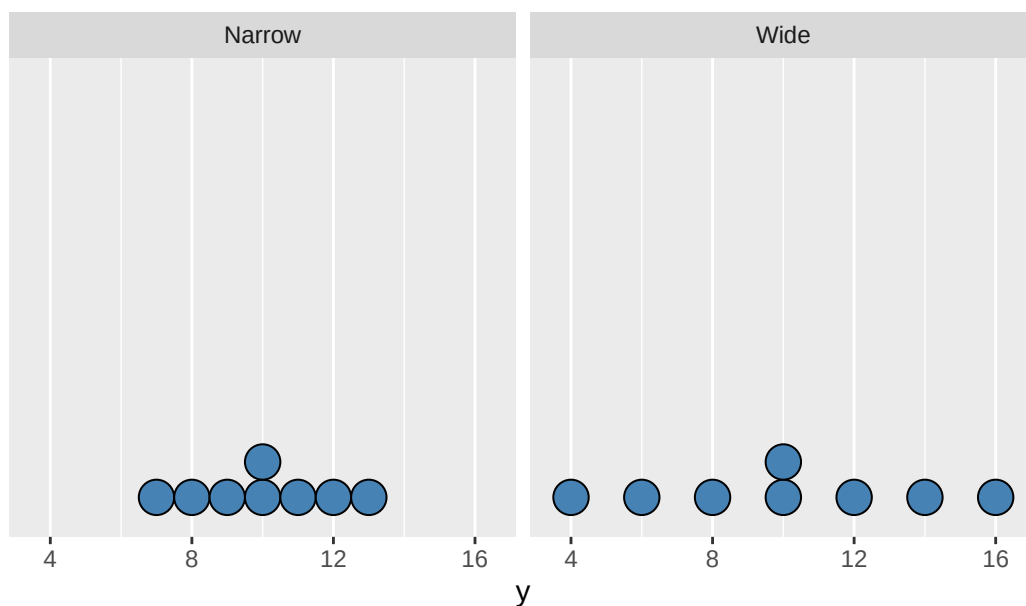


```
ggplot(compare, aes(x = y, fill = group)) +
  geom_histogram(binwidth = 1, position = "identity", alpha = 0.5, color = "white") +
  labs(title = "Histograms overlaid", x = "y", y = "Count")
```



```
ggplot(compare, aes(x = y)) +
  geom_dotplot(binwidth = 1, method = "histodot", fill = "steelblue") +
  facet_wrap(~ group) +
  labs(title = "Dot plots by group", x = "y") +
  scale_y_continuous(NULL, breaks = NULL)
```

Dot plots by group



group_by() descriptives and ggplot with fill

Check in R above already used `group_by(group) |> summarize(...)` on `compare` and mapped `fill` / `facet_wrap` in `ggplot2`. The same pattern — one numeric variable summarized within each level of a grouping variable — is the heart of comparing groups descriptively, and it is the main R skill you need for Project 3.

Syntax note. As in Section 2, `summarize()` and `summarise()` are equivalent, and `|>` matches `%>%` for piping; this lecture uses `|>` and `summarize()`.

A compact per-group summary of `compare`:

```
compare |>
  group_by(group) |>
  summarize(
    n = n(),
    mean_y = mean(y),
    sd_y = sd(y),
    median_y = median(y)
  )
```

```
# A tibble: 2 x 5
  group      n mean_y sd_y median_y
<fct> <int> <dbl> <dbl>   <dbl>
1 Narrow     8     10     2     10
2 Wide       8     10     4     10
```

💡 Interpret the output — code comes in the demo lab

The chunks below use tidyverse to produce tables and figures. Focus on reading counts, proportions, summaries, and graphs. You will learn how to write these steps in [Demo 1](#) and [Demo 2](#).

Example 1 — Flipper length by species

```
library(tidyverse)
if (!requireNamespace("palmerpenguins", quietly = TRUE)) {
  install.packages("palmerpenguins", repos = "https://cloud.r-project.org")
}
library(palmerpenguins)

penguins |>
  group_by(species) |>
  summarize(
    n = n(),
    median_flip = median(flipper_length_mm, na.rm = TRUE),
    iqr_flip = IQR(flipper_length_mm, na.rm = TRUE),
    .groups = "drop"
  )
```

```
# A tibble: 3 x 4
  species      n median_flip iqr_flip
<fct>    <int>      <dbl>   <dbl>
1 Adelie   152      190      9
2 Chinstrap  68      196     10
3 Gentoo  124      216      9
```



```
ggplot(penguins, aes(x = species, y = flipper_length_mm, fill = species)) +
  geom_boxplot() +
  labs(title = "Flipper length by species", x = "Species", y = "Flipper length (mm)") +
  theme(legend.position = "none")
```

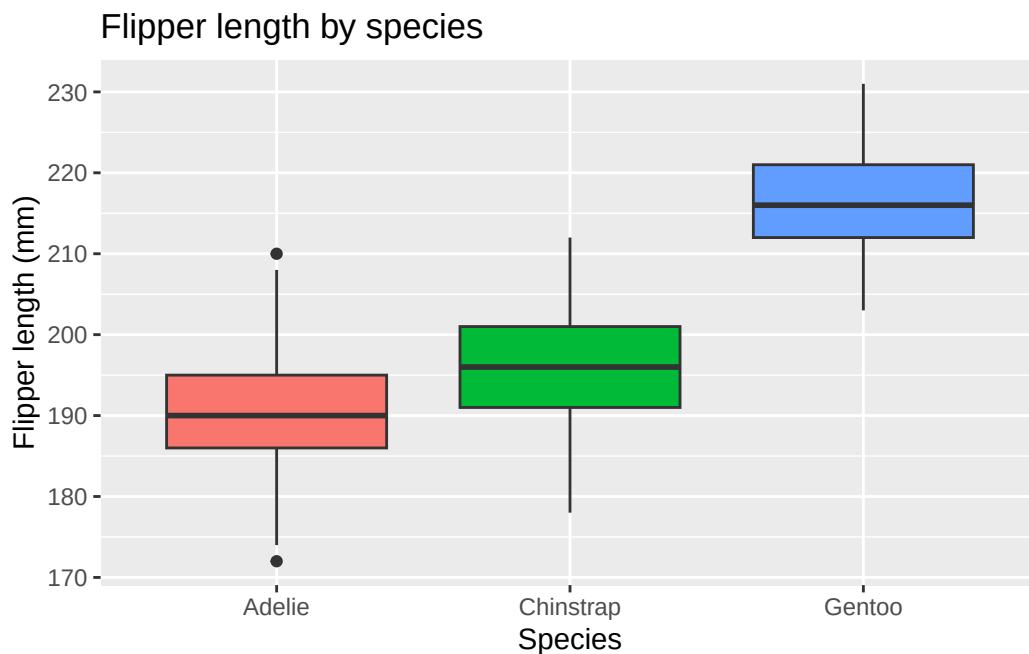


Figure 13: Flipper length differs clearly across species.

Interpret: Which species has the highest typical flipper length (compare medians)? Which has the widest middle 50% (IQR)?

Answer: Gentoo has the highest typical flipper length (median ≈ 216 mm, vs ≈ 190 mm for Adelie and ≈ 196 mm for Chinstrap). The three IQRs are similar (≈ 9 – 10 mm), so no species' middle 50% is wider than the others.

Example 2 — Body mass by sex

```
penguins |>
  filter(!is.na(sex)) |>
  group_by(sex) |>
  summarize(
    median_mass = median(body_mass_g, na.rm = TRUE),
    iqr_mass = IQR(body_mass_g, na.rm = TRUE),
    .groups = "drop"
  )
```

```
# A tibble: 2 x 3
  sex    median_mass iqr_mass
<fct>      <dbl>     <dbl>
1 female    3650      1200
2 male     4300     1412.
```

```
ggplot(filter(penguins, !is.na(sex)), aes(x = sex, y = body_mass_g)) +
  geom_boxplot(fill = "tan") +
  labs(title = "Body mass by sex", x = "Sex", y = "Body mass (g)")
```

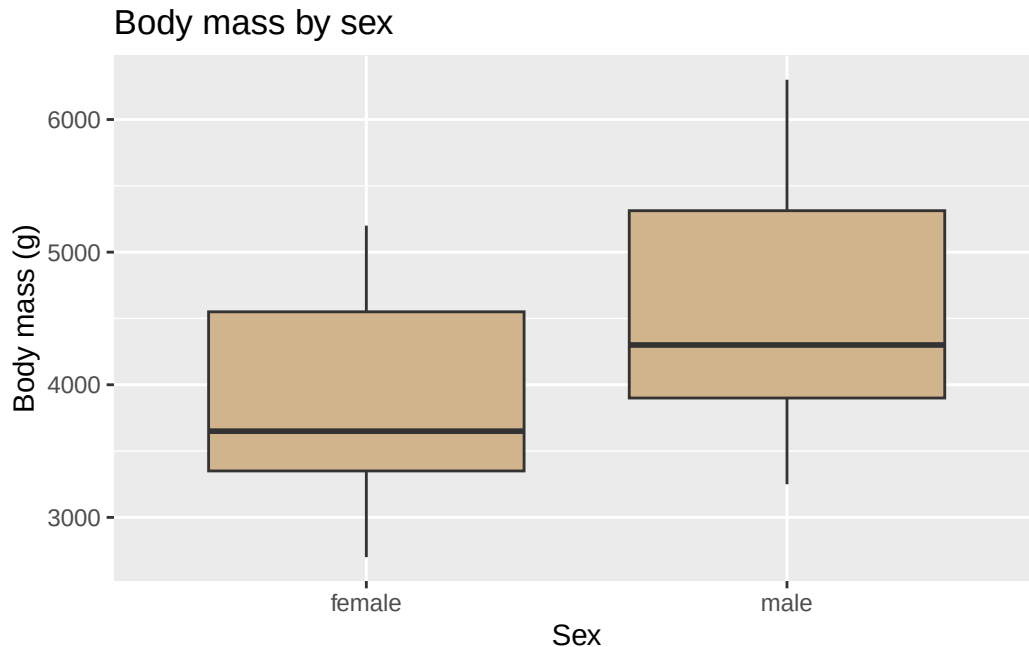


Figure 14: Body mass by sex — compare box positions and spread.

Interpret: One sentence comparing center and spread between groups.

Answer: Males have a higher typical body mass (median ≈ 4300 g vs ≈ 3650 g for females) and a somewhat larger spread (wider box).

Example 3 — Highway MPG by drive type

```
data("mpg", package = "ggplot2")
mpg |>
  group_by(drv) |>
  summarize(median_hwy = median(hwy), iqr_hwy = IQR(hwy), .groups = "drop")
```

```
# A tibble: 3 x 3
  drv   median_hwy iqr_hwy
<chr>     <dbl>   <dbl>
1 4         18       5
2 f         28       3
3 r         21       7
```

```
ggplot(mpg, aes(x = drv, y = hwy, fill = drv)) +
  geom_boxplot() +
  labs(title = "Highway MPG by drive type", x = "Drive type", y = "Highway MPG") +
  theme(legend.position = "none")
```

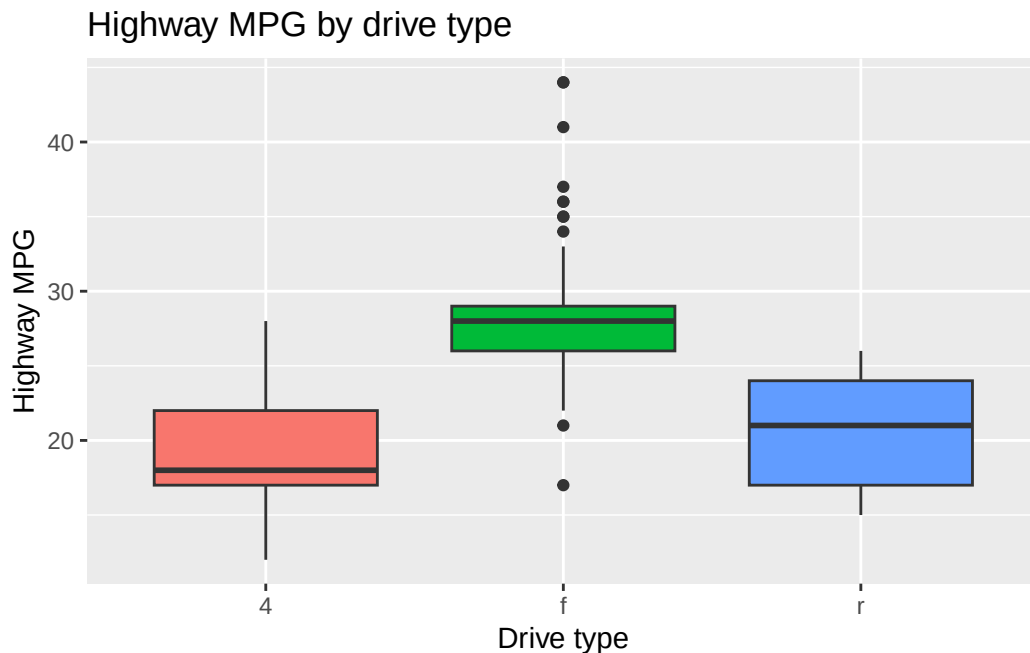


Figure 15: Highway MPG by drive type (r, f, 4).

Interpret: Which drive type has the highest median highway MPG?

Answer: Front-wheel drive (f) has the highest median highway MPG, well above rear-wheel (r) and four-wheel (4) drive.

i From this sample to a population — inference reference (optional)

Described here: the means (or medians) of a numerical outcome across the levels of a categorical variable.

Population parameters: the group means $\mu_1, \mu_2, \dots, \mu_k$ — the true means within each of the k groups.

Two groups ($k = 2$): two-sample t -test. If you want to **compare two population means** ($H_0 : \mu_1 = \mu_2$), then — after checking conditions (independent groups; each group large or roughly symmetric) — use the two-sample (Welch) t -statistic

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}},$$

which compares the observed difference in sample means to its standard error; Welch's version does not assume equal group SDs. Software returns the p -value, the degrees of freedom, and a confidence interval for $\mu_1 - \mu_2$.

Three or more groups ($k \geq 3$): one-way ANOVA. With more than two categories you do **not** run many t -tests (that inflates the error rate). Instead test whether **all** the group means are equal,

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k \quad \text{vs.} \quad H_a : \text{at least one } \mu_i \text{ differs,}$$

using the F -statistic, which is a ratio of variability between the group means to variability within the groups,

$$F = \frac{\text{between-group variability (MS}_{\text{between}})}{\text{within-group variability (MS}_{\text{within}})}.$$

If the groups truly have the same mean, F is near 1; a large F (small p -value) says at least one group mean differs — though ANOVA alone does not tell you *which* one. It is the natural extension of the two-sample t -test from 2 groups to k groups (and with $k = 2$ the two procedures agree). Let software do the work:

```
t.test(y ~ group)           # two groups: difference in means  $\mu_1 - \mu_2$  (CI + p-value)
oneway.test(y ~ group)      # 3+ groups: one-way ANOVA (do any group means differ?)
# summary(aov(y ~ group))   # classic one-way ANOVA table
```

Both are introduced in **Module 4** (two-group comparison; one-way ANOVA extends it to several groups).

Check yourself — One numerical and one categorical

1. Dataset W: 4, 6, 8, 10, 10, 12, 14, 16. Both N and W have mean 10. Which has larger SD?
2. In tidyverse, which two verbs together produce one summary row per group?
3. Which ggplot2 geometry compares a numerical variable across groups?

Answers — One numerical and one categorical

1. W — values are spread farther from the mean.
2. `group_by()` followed by `summarize()`.
3. `geom_boxplot()` with the categorical variable on x (or y) and the numerical variable on the other axis.

i Methods Map Reference Tables:

- [Organizing the full statistics toolbox](#) — descriptive vs. inferential map